

MECHENG726: Acoustics for Engineers

Chapter 5: Microphones & Loudspeakers (ver1.0)

Learning outcomes

By the end of these lecture notes, students will understand

- ▶ How loudspeakers convert a signal to sound.
- ▶ How to read specs of a loudspeaker.
- ▶ How microphones pick up sound and convert it to a signal.
- ▶ Different types of microphones and their classifications.
- ▶ How to read specs of a microphone.
- ▶ Advanced techniques using microphones and loudspeakers.

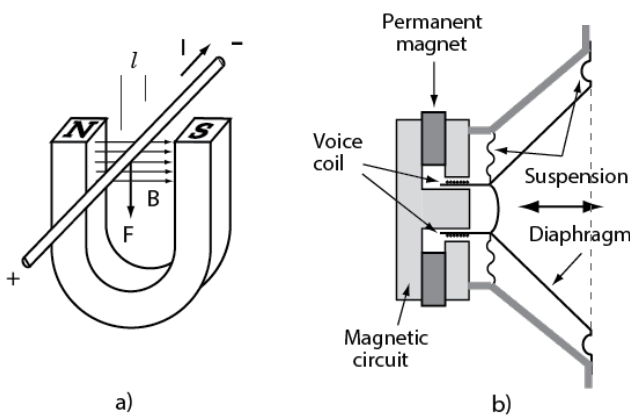
Section 1: Loudspeakers

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Loudspeaker: how it works

- ▶ Transforming electrical signals into acoustic sound is realised using **loudspeakers** or **headphones**.
- ▶ A loudspeaker consists of one or more **driver** unit(s), which is/are normally accommodated in an enclosure.
- ▶ Most driver units are based on electro-dynamic principle.



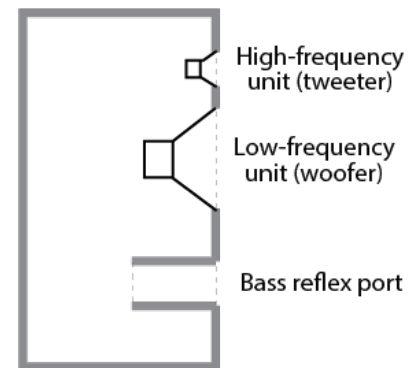
Pulkki, Fig.4.1

a) If a wire in a magnetic field of strength B carries a current I , a mechanical force $F=BIl$ acts to move the wire. l is the effective length of wire in the magnetic field.

b) A cylinder-shaped **voice coil** can move in the air gap of a permanent magnet with a strong magnetic field. A voltage from a power amplifier applied to the coil results in a current through the coil, causing a force to the coil. A cone-shaped **diaphragm** attached to the coil vibrates and make the air more to generate the sound wave.

Loudspeaker: how it works

- ▶ The driver being used itself is not enough for good sound reproduction especially for low frequency sound.
- ▶ Normally a driver is attached to an enclosure that resonates at low frequencies. The enclosure has two types:
 - ▶ **Closed box** has no holes other than that for the driver
 - ▶ **Bass-reflex** has a tube for low frequency sound to propagate
- ▶ More than one drivers of different size are often attached to the same enclosure to cover broader frequency range.
 - ▶ 2-way loudspeaker has **tweeter** & **woofer**.
 - ▶ 3-way loudspeaker has **tweeter**, **mid-range**, & **woofer**.



Pulkki, Fig.4.1

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Loudspeaker: how it works

- ▶ **Power amplifier** is always needed to drive the driver of a loudspeaker (need to incur voltage to get a current).
- ▶ Some loudspeakers come with its power amplifier hosted inside its enclosure which tend to be heavier.



2-way buss-reflex loudspeaker **with**
power amp (Genelec 8020C)



2-way buss-reflex loudspeaker **without**
power amp (JBL control monitor)

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Specs

- ▶ There are several terminologies utilised to describe the specs of a loudspeaker.
 - ▶ **Frequency response** illustrates how faithfully the loudspeaker will reproduce the sound in terms of its spectral shape. It is typically represented by a plot of the amplitude response. Be aware that the frequency response of the power amplifier connected to the loudspeaker also affects the quality of the played sound.
 - ▶ **Frequency range** refers to how low and high the loudspeaker can reproduce sound with certain amount of energy. The range often comes with maximum deviation from 0 dB e.g. ± 3 dB.
 - ▶ **Power handling** indicates how much power the loudspeaker can bear without causing any damage. Displayed either in r.m.s. or peak values in watts. **Impedance** is also related to this spec.
 - ▶ **Dimension** and **weight** are also a part of specs.

Example product

- ▶ Genelec 8020D
 - ▶ <https://www.genelec.com/8020d>

Technical Specifications



SPL
100 dB



Frequency Response
56 Hz - 25 kHz (± 6 dB)



Driver Dimensions
 $\varnothing$ 105 mm Bass + $\varnothing$ 19 mm Treble (view in inches)



Weight
3.2 kg / 7.1 lb



Amplifier Power
50 W Bass (Class D) + 50 W Treble (Class D)



Accuracy of Frequency Response
 ± 2.5 dB (62 Hz - 20 kHz)



Dimensions
H 242 x W 148 x D 142 mm, with Iso-Pod™ (view in inches)



Connections
1 x XLR Analog Input

Example product

► Datasheet:

https://assets.ctfassets.net/4zjzn055a4v/6TJzRxdYXeEyYcMoMCIQMi/ac9a1702e8e0b8a9ddb671c288605c42/8020d_opman_en_fi.pdf

GENELEC®

8020D



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Section 2: Microphones

Microphone: a transducer

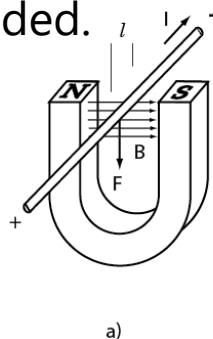
- ▶ A microphone is a transducer that transforms a **sound wave** (pressure) propagating in the air into an **electrical signal** (voltage).
- ▶ There are several **conversion principles** discussed in the following slides.

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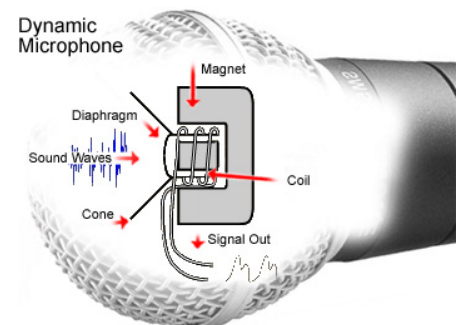
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Microphone: how it works

- ▶ **Dynamic microphone** is based on the electrodynamic principle used for the dynamic loudspeaker, but the direction of the conversion is opposite.
- ▶ A diaphragm moves due to the medium's (air) vibration and causes voltage across the voice coil that is proportional to the velocity of the diaphragm.
- ▶ No power supply is needed.



Pulkki, Fig.4.1



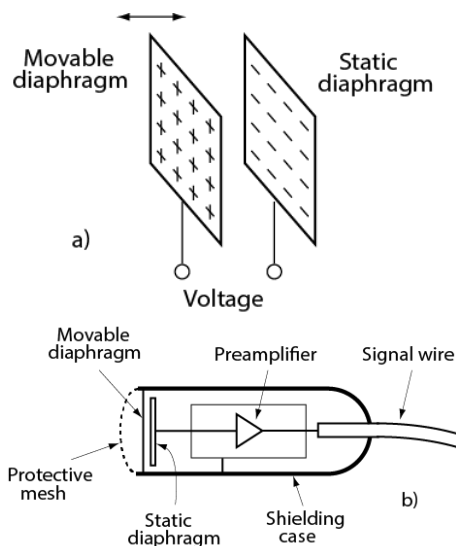
http://www.songsofthecosmos.com/encyclopedia_of_modern_music/D/dynamic_microphone.html

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Microphone: how it works

- ▶ **Condenser microphone** uses the properties of electret condensers (i.e. capacitors).
- ▶ Power supply is needed to incur the voltage across the electrodes of the condenser.



- The sound pressure variations make one electrode of the condenser move while the other is kept fixed.
- If an electric charge is present between the electrodes, the change in distance between them will change the voltage between them too.
- This tiny difference of voltage change is amplified and sent as the audio signal.

$$Q = CV$$

$$C = \frac{\epsilon A}{d}$$

A: area of diaphragm
d: distance between diaphragm
 ϵ : permittance

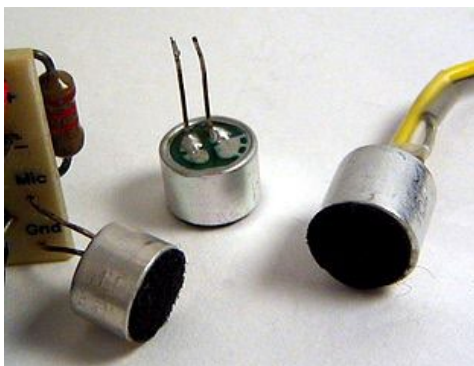
Pulkki, Fig.4.2

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Microphone: how it works

- ▶ **Electret microphone** is a variant of condenser microphone.
- ▶ It uses electret material, which has electric charge trapped within, between the electrodes.
- ▶ Electret microphones have wide spread use especially in portable devices, e.g. smartphones, laptop PCs.



https://upload.wikimedia.org/wikipedia/commons/thumb/b/5/57/Electret_condenser_microphone_capsules.jpg/312px-Electret_condenser_microphone_capsules.jpg

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Microphone: how it works

- ▶ **MEMS** (MicroElectrical-Mechanical System) **microphone** is a new paradigm of microphone design.
- ▶ A pressure-sensitive diaphragm is etched directly into a silicon wafer by MEMS processing techniques, and is usually accompanied with integrated preamplifier.
- ▶ **Digital MEMS microphones** have built in A/D converter on the same chip making the chip a digital microphone which can be integrated with modern digital products.
- ▶ It is very **tiny** so it would be suitable for mobile devices and large size microphone arrays*.



*We cover this in the following section.

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http://www.eetimes.com/author.asp?section_id=36&doc_id=1286564

Classification by principle

- ▶ Pros & Cons, and applications

	Dynamic	Condenser	MEMS
Pros	• No power supply needed • Robust • Cheap	• High sensitivity • High quality	• Very small • Robust
Cons	• Low sensitivity • Low quality • Can be large	• Power supply needed (except electret type) • Vulnerable to vibration/humidity • Could be expensive	• Expensive but similar to condenser mics
Application	• Outdoor use • Public address system/concert	• Indoor use • Computer/electronic devices	• Mobile devices • Microphone arrays

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Specs

- ▶ There are several terminologies utilised to describe the specs of a microphone.
 - ▶ **Sensitivity** expresses what voltage a microphone will produce at a certain sound pressure level. Utilised to pick up small sound while being less affected by noise. Measured in mV per Pascal at 1 kHz.
 - ▶ **Dynamic range** is the ratio of the amplitude of the loudest possible undistorted sine wave to the root mean square noise amplitude.
 - ▶ **Frequency response** illustrates whether the microphone will pick up sound faithfully or will introduce colouration to the spectrum.
 - ▶ **Directivity** refers to the sensitivity of a microphone to the same sound when it arrives from different angles.
 - ▶ **Signal-to-noise ratio (SNR)** measures how much desired sound is present in an audio recording, as opposed to noise. Noise in the context of microphone's spec is typically the internal noise.

Specs (cont'd)

- ▶ **Input impedance** refers to electronical impedance of the microphone. Lower impedance is better as the input is less affected by the devices/cables connected to the microphone.
- ▶ **Maximum SPL** is the maximum sound pressure level that the microphone can take without distortion.
- ▶ A good microphone should cover a wide range of frequencies, e.g. 20 – 20 kHz, with a flat frequency response, and a wide dynamic range. It should also have high sensitivity, cause low distortion and noise, and have a desired directivity pattern.

Microphone: Classifications

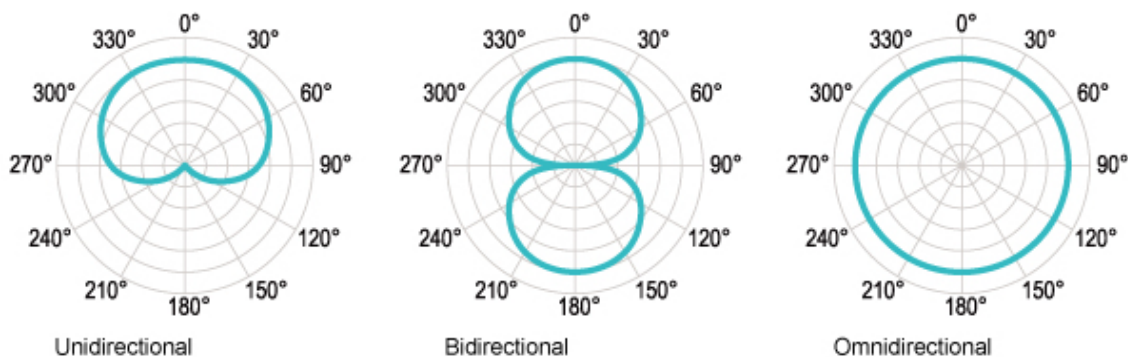
- ▶ Various types of microphones are available in the market.
- ▶ They can be classified by different aspects. We will classify microphones into different categories from the following point of view.
 - ▶ Principle
 - ▶ Directivity
 - ▶ Application and purposes

Classification by principle

- ▶ As we have seen earlier, microphones are based on various different principles.
- ▶ One way of classifying microphones is using such difference in their mechanisms including
 - ▶ Dynamic microphone
 - ▶ Condenser microphone
 - ▶ MEMS microphone

Classification by directivity

- ▶ Directivity specifies a microphone's ability to spatially discriminate sound arriving from different angles.
- ▶ **Polar patterns** are often used to show the directivity of a microphone. The radius of the pattern shows the gain (sensitivity) of the microphone to that specific angle.
- ▶ Normally polar pattern changes at different frequency.



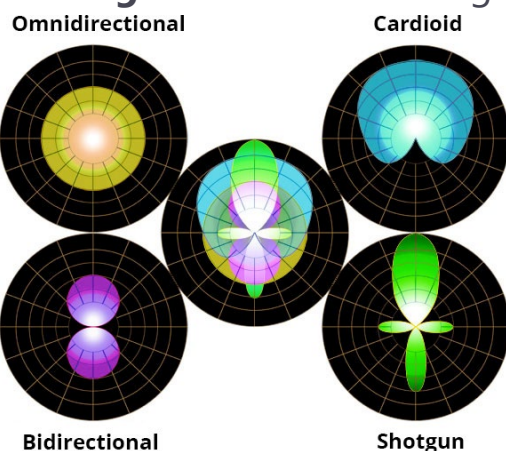
http://www.yamahaproaudio.com/global/en/training_support/selftraining/pa_guide_beginner/microphone/

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Classification by directivity (cont'd)

- ▶ Several types are available for directivity including:
 - ▶ **Omni-directional** has its sensitivity being identical in every angle.
 - ▶ **Uni-directional (or Cardioid)** has high sensitivity to forward while pointing a null to the back.
 - ▶ **By-directional** has two separate region where its sensitivity is high.
 - ▶ **Shotgun** has a narrow angular region where its sensitivity is high.



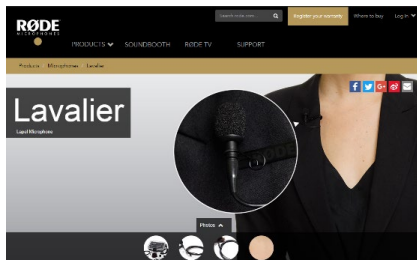
<https://www.bhphotovideo.com/explora/audio/buying-guide/camera-microphones>

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Classification by purposes (cont'd)

- ▶ There are various types of microphones for different purposes.
 - ▶ **Lapel** microphone is a very small microphone that can be worn by a speaker. Typically it has omni-directional or cardioid directivities.
 - ▶ **Boundary** microphone is mounted on a flat surface (e.g. table) and typically has an omni-directional directivity. Used for meetings.
 - ▶ **Wireless** microphone transmits the recorded signal to the amplifier via radio waves.



Lapel microphone¹



Boundary microphone²



Wireless microphone³

¹<http://www.ode.com/microphones/Lavalier>

²https://commons.wikimedia.org/wiki/File:Audio-technica_ATM87R.jpg

³<http://www.shure.com/americas>

Classification by purposes

- ▶ **Hydrophone** is a microphone designed specifically for recording underwater sound.
- ▶ **Stereo** microphone consists of two uni-directional microphones each of which captures sound from left or right direction.
- ▶ **Bone conduction** microphone picks up sound conducted through the bones of the skull to the inner ear. Robust to noisy environment.



Hydrophone¹



Stereo²



Bone conduction³

¹https://en.wikipedia.org/wiki/Hydrophone#/media/File:Hydrophone_being_lowered_into_the_Atlantic.jpg

²<https://www.zoom-na.com/products/field-video-recording/field-recording/h6-handy-recorder/capsules#xy>

³https://commons.wikimedia.org/wiki/File:Goldendance_bone_conduction_headset.jpg

Power supply to microphones

- ▶ Some types of microphones (e.g. condenser) require power supply, which can be provided by **batteries** or **external power supply**.
- ▶ Power from external supply is sent through the cable as the audio signal is transmitted. A few protocols are available for the power supply:
 - ▶ **Plug-in power** supply provides 5-9 V DC power supply.
 - ▶ **Phantom power** supply provides 48 V DC power supply.

Microphone power amplifier

- ▶ Signals a microphone can generate are generally very tiny, so a **power amplifier** is needed to amplify the signal.
- ▶ Because the tiny signal from the microphone is vulnerable to noise, the length of cable between the microphone and power amplifier should be minimised.
- ▶ Some microphone has an amplifier attached to the connector or even the microphone capsule itself.
- ▶ An appropriate amplifier should be selected as the amplification properties of the power amplifier also affects the overall quality of the recorded signal. Class A amplifier is often preferred for high fidelity audio recording.

Connectors

- ▶ Various types of connectors are available for connecting a microphone to a power amplifier.
 - ▶ **XLR (a.k.a. Cannon)** has 3 to 7 pins. It can send phantom power supply to the microphone through cable. This type is often used for professional use.
 - ▶ **1/4 inch phone** is widely used both for professional and consumer uses. It is also used for connecting loudspeakers/headphones.
 - ▶ **3.5 mm/2.5 mm** is often used for plug-in power type microphones such as PC. It has both **mono/stereo** types.



Examples datasheets

- ▶ Rode Lavalier
 - ▶ http://cdn1.ode.com/lavalier_datasheet.pdf
- ▶ Rode NTG-1
 - ▶ http://cdn1.ode.com/ntg1_datasheet.pdf
- ▶ Rode NT6
 - ▶ http://cdn1.ode.com/nt6_datasheet.pdf

Section 3: Measuring microphones & loudspeakers

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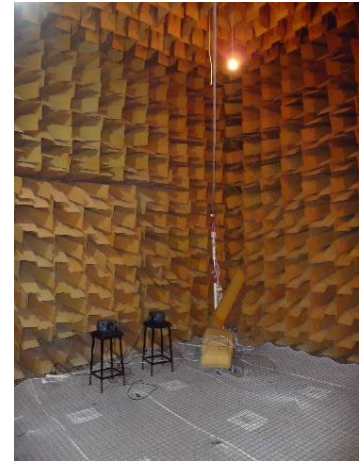
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Why do we need to measure mics/spks?

1. To know the actual characteristics:
Actual characteristics of microphones/loudspeakers would never be the same as that stated in their specs and can only be known by practically measuring the devices.
2. To measure other objects:
Characteristics of microphones/loudspeakers are included in the measurements of other objects, e.g. rooms, which should be removed in order to isolate the characteristics of the targeted objects.

Measurement environment

- ▶ Measurement of microphones/loudspeakers normally takes place in an **anechoic chamber** in order to minimise the effect of room's properties in the measurement.
- ▶ The walls are covered with highly absorptive material which practically eliminates all reflections and reverberations so that only direct sound from a source to a point can be measured.
- ▶ An anechoic chamber is also acoustically insulated so the noise level inside the chamber is extremely low.



Anechoic chamber at Acoustics Lab of University of Auckland

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Instrumentation microphone

- ▶ Since acoustic measurement normally cannot avoid using a microphone, a microphone with "ground truth" characteristics is required.
- ▶ **Instrumentation (measurement) microphones** should have a flat frequency (amplitude) response so that spectral colouring caused by the microphone will be minimised.
- ▶ In order to make sure the sensitivity follows the spec value, **calibration** is conducted using a calibrator.



Instrumentation microphone¹



Calibrator²

¹<http://www.gras.dk/46ae.html>

²<http://www.gras.dk/42ag.html>

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Frequency response measurement

- ▶ Frequency response of a microphone/loudspeaker/room can be given by measuring the **impulse response** of the object (covered in Chapter 4).
- ▶ **FFT** (covered in Chapter 2) is normally used to convert the impulse response to the frequency response.
- ▶ A calibrated instrumentation microphone is used for the measurement of loudspeakers.



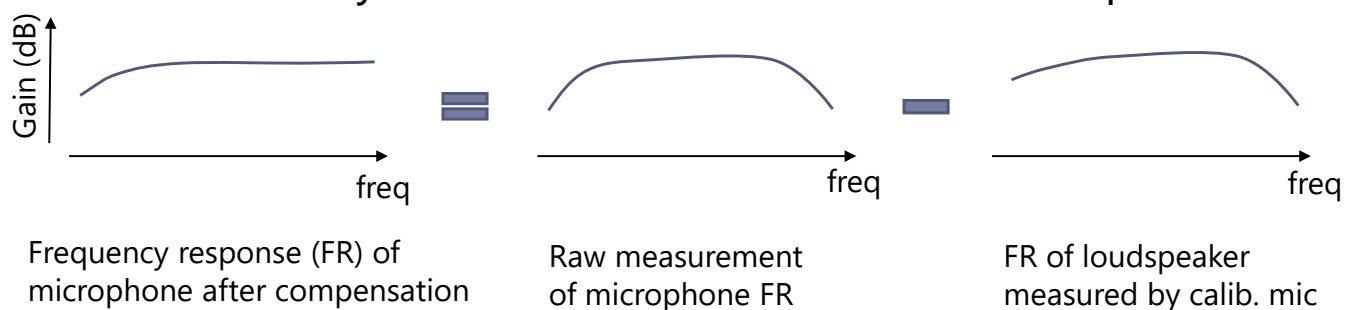
Measuring a loudspeaker using a calibrated instrumentation microphone.

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Frequency response measurement

- ▶ For microphones, an arbitrary loudspeaker is used to play the swept sine signal for impulse response measurement.
- ▶ The raw measurement will also include the loudspeaker's characteristics which can be compensated (removed) by post-processing.
- ▶ The compensation can be realised by subtracting (in dB) the frequency (amplitude) response of the loudspeaker measured by a calibrated instrumentation microphone.

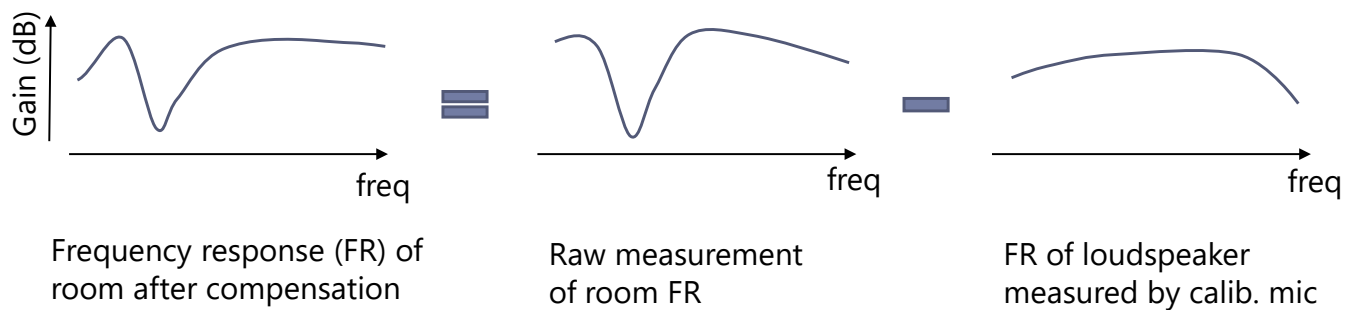


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Frequency response measurement

- ▶ For (acoustics of) rooms, same as the loudspeaker, raw measurement of room impulse response is taken by a calibrated instrumentation microphone first, which is converted to the frequency response by FFT.
- ▶ The frequency response of the loudspeaker (measured by instrumentation microphone) is then removed from the raw measurement of the room.

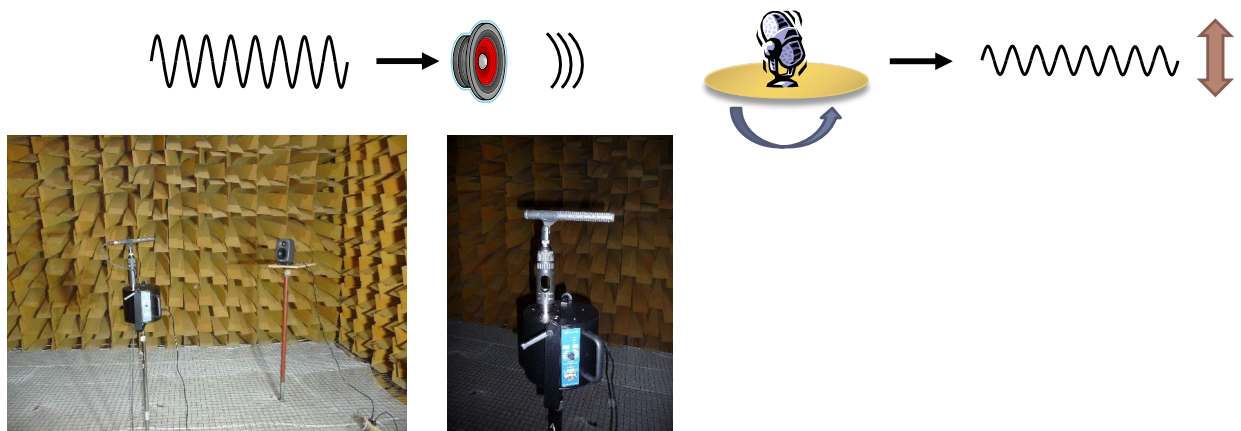


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Directivity pattern

- ▶ Directivity pattern (polar plot) is measured by playing a sine wave while the device is rotated at a constant speed.
- ▶ Variation of the amplitude of the recorded sine wave is plotted as directivity pattern.
- ▶ Turntable is used for rotating the device.



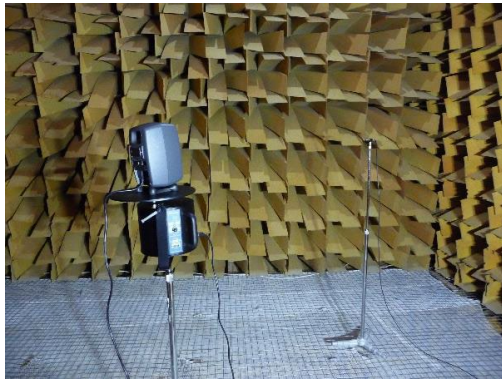
Measuring directivity patterns of a microphone placed on a turn table.

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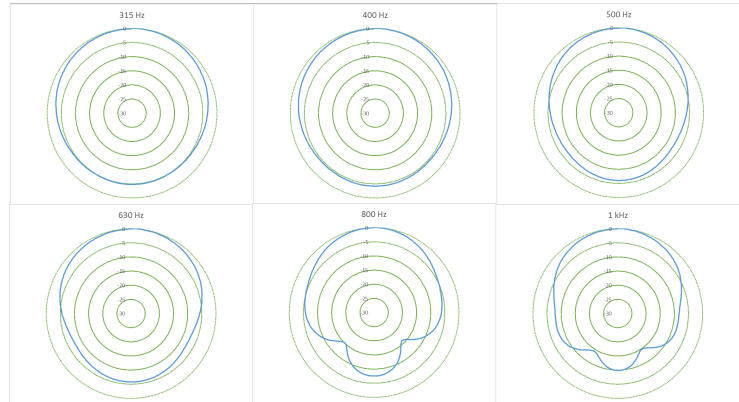
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Directivity pattern

- ▶ Directivity pattern is measured at different frequency by changing the frequency of the sine wave played.
- ▶ Typically frequencies at 1/3 octave band are selected.



Measuring directivity patterns of a loudspeaker placed on a turn table.



Measured directivity patterns at different frequencies.

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Section 4: Advanced techniques using microphones/loudspeakers

Microphone arrays

- ▶ **Microphone array** is a set of more than one microphone that is able to detect/utilise spatial information (location) of sound sources.
- ▶ Typical applications of microphone arrays are:
 - ▶ **Emphasising** sound sources located in particular direction/location
 - ▶ **Estimating direction/location** of sound sources

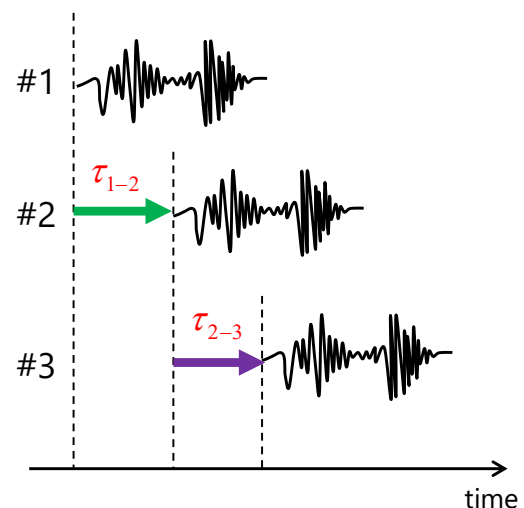
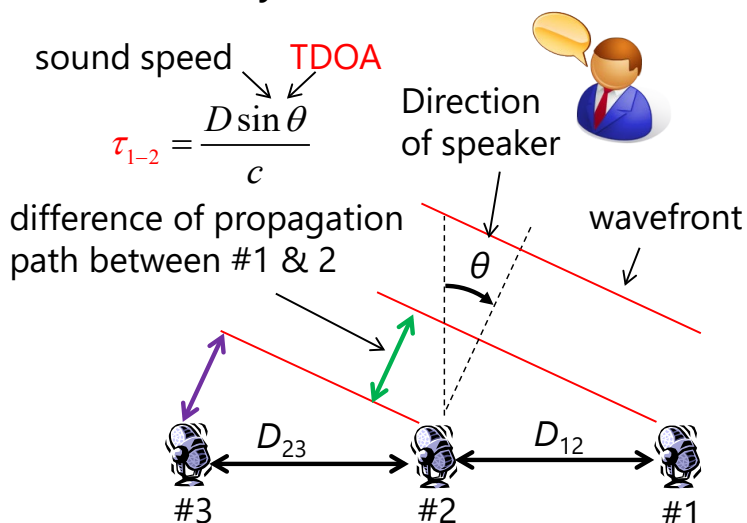
which are realised by digital signal processing algorithms applied to the signals observed by the microphone array.



Microphone array using 4 lapel microphones

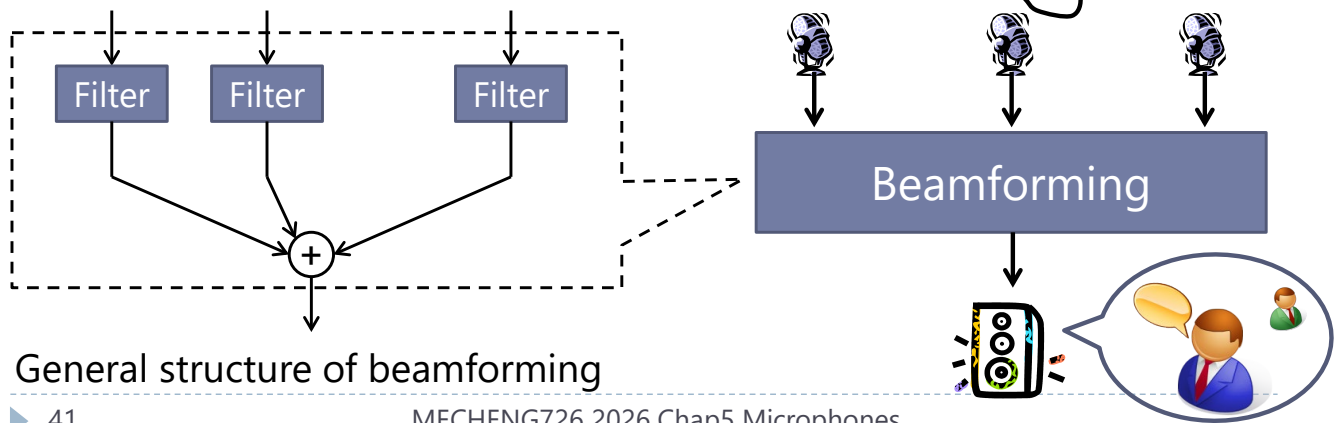
Microphone arrays

- ▶ Sound wave arrives each microphone at different timing causing **time difference of arrival (TDOA)**.
- ▶ TDOA is a function of the angle of the sound source w.r.t. the array.



Beamforming

- ▶ Signal processing technique that forms a directivity pattern
- ▶ Consists of FIR filters connected to each microphone and an adder
- ▶ Directivity can be electronically varied by changing the property of the filters.



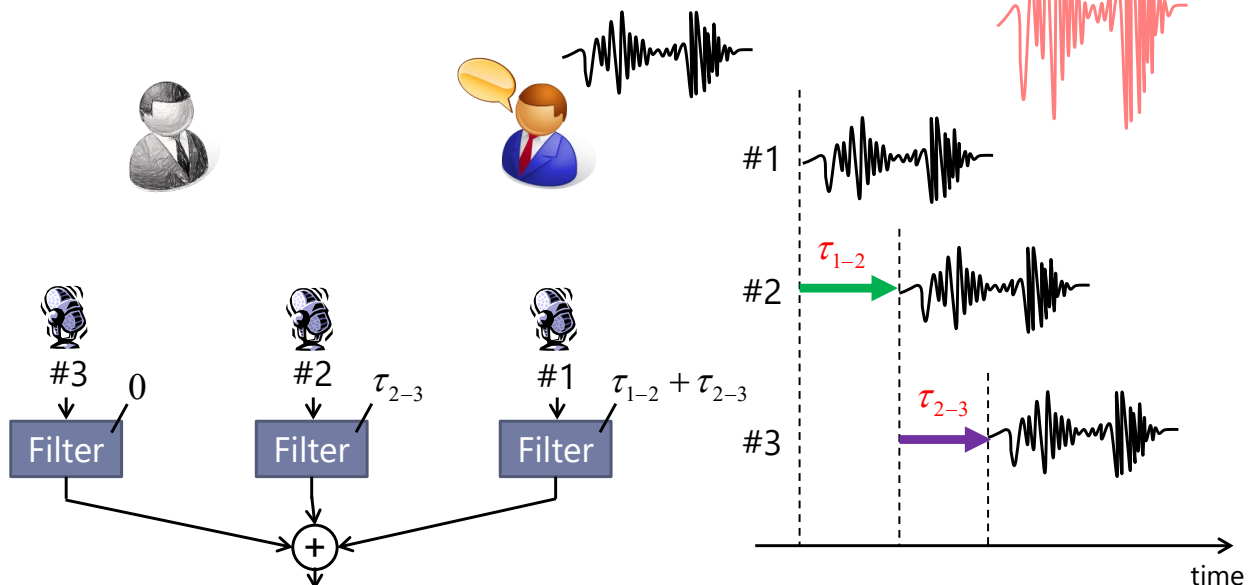
General structure of beamforming

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Beamforming

- ▶ Delay-sum beamformer is the simplest technique.
- ▶ The FIR filters are designed to delay the signals so that the target signal is aligned across the microphones.

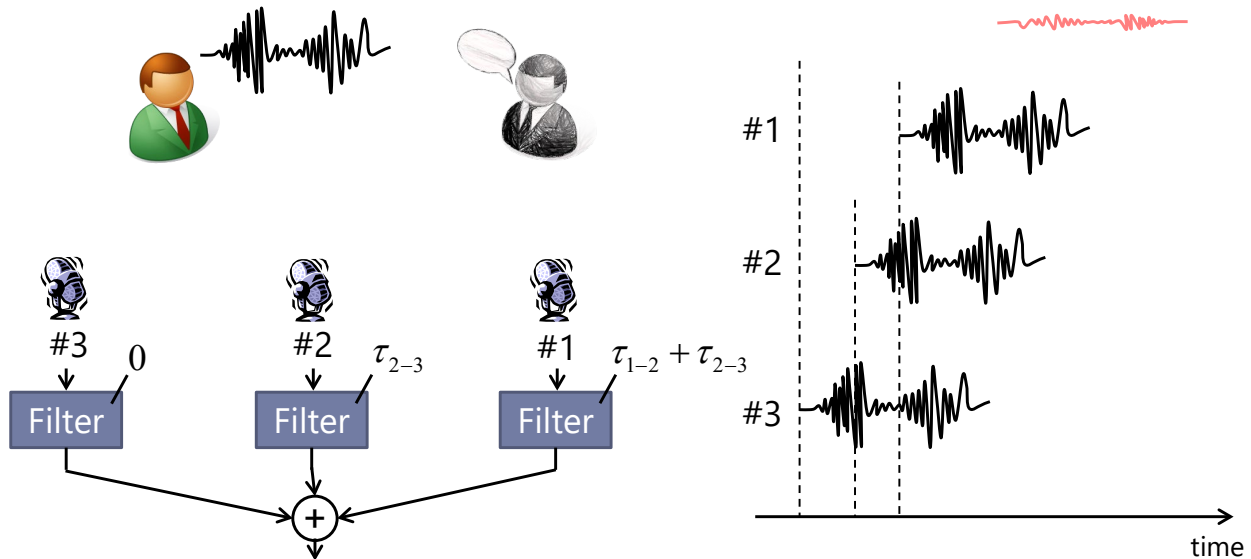


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Beamforming

- ▶ **Delay-sum beamformer** is the simplest technique.
- ▶ The FIR filters are designed to **delay** the signals so that the target signal is aligned across the microphones.

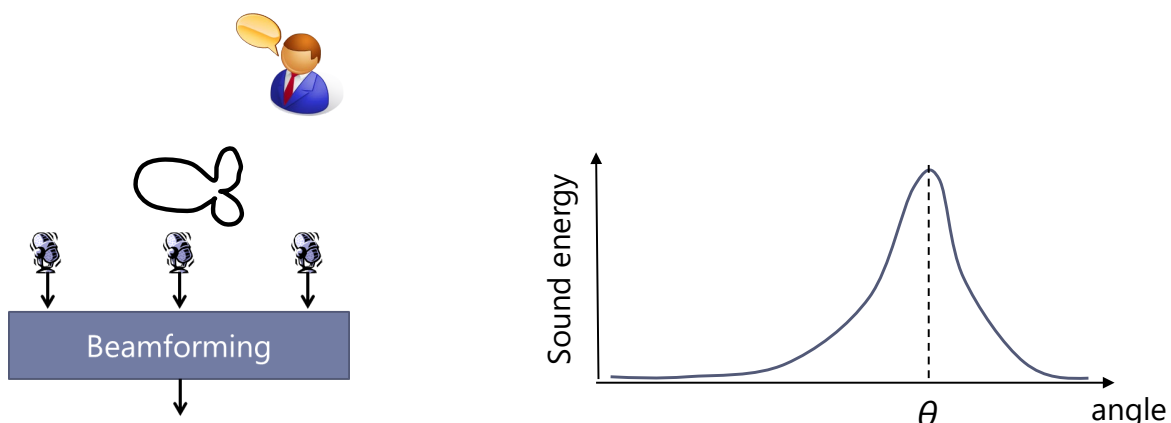


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Source localisation

- ▶ Location/direction of sound sources can be estimated using signal processing techniques applied to microphone array observations.
- ▶ **Steered beamformer method** estimates source direction by measuring sound energy arriving from different angles.



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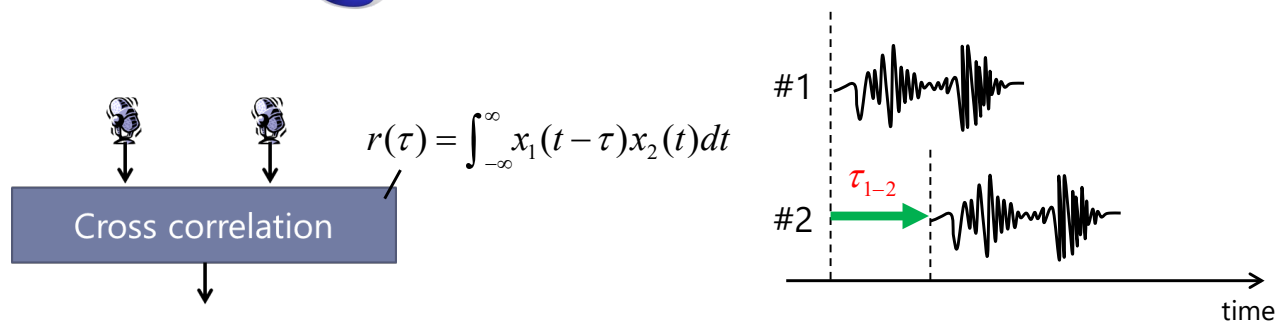
Source localisation

- ▶ **Cross correlation method** calculates the TDOA by comparing the measurements between two microphones.
- ▶ The angle of the source is calculated from the estimated TDOA.

$$\tau_{1-2} = \frac{D \sin \theta}{c} \quad \Rightarrow \quad \theta = \sin^{-1} \left(\frac{c \tau_{1-2}}{D} \right)$$

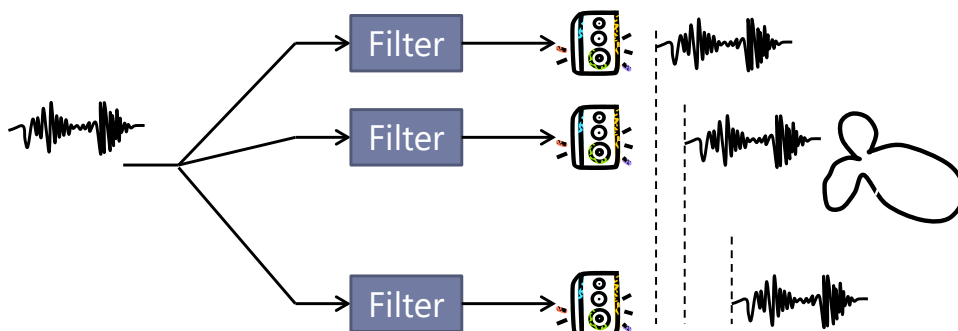


$$\tau_{1-2} = \arg \max_{\tau} \int_{-\infty}^{\infty} x_1(t - \tau) x_2(t) dt$$



Loudspeaker arrays

- ▶ An array of loudspeakers is also used for applications in acoustics.
- ▶ The most common purpose of using a loudspeaker array is to realise a sharp directivity pattern.
- ▶ Sound source is emitted to a particular angle by adjusting the phase of the signal between the loudspeakers. Same mechanism as beamforming for microphone arrays.



Practical example

- ▶ **Acoustic camera** is an imaging device used to locate sound sources and characterise them using a microphone array with beamforming.



¹<https://www.acoustic-camera.com/en/products/microphone-arrays.html>

²<https://www.acoustic-camera.com/en/products/microphone-arrays/ring-48-75-ac-pro.html>

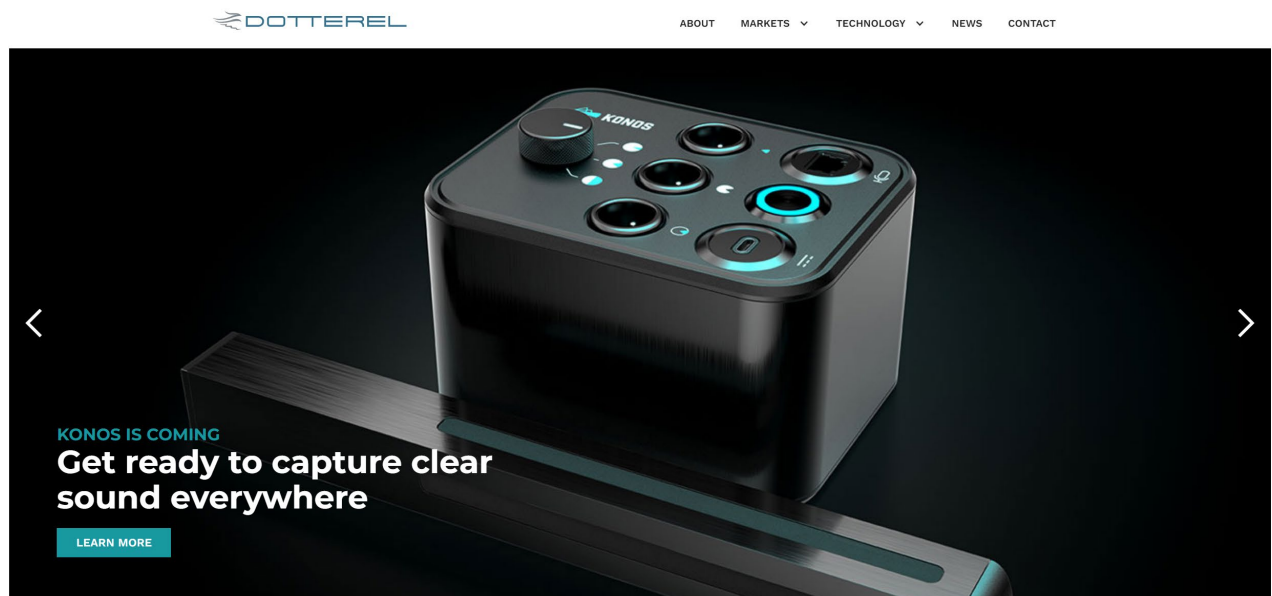


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Practical example

- ▶ Microphone array



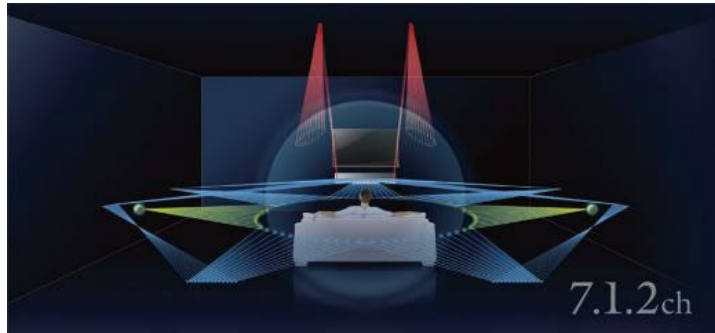
<https://www.konos-sound.com/product>

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Practical example

- ▶ Loudspeaker array for surround sound system



https://au.yamaha.com/en/products/audio_visual/sound_bar/ysp-5600/features.html#product-tabs

Loudspeaker array used for 3D surround sound system.
YAMAHA YSP-5600

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Practical example

- ▶ Virtual sound reproduction



Restaurant / Cafe

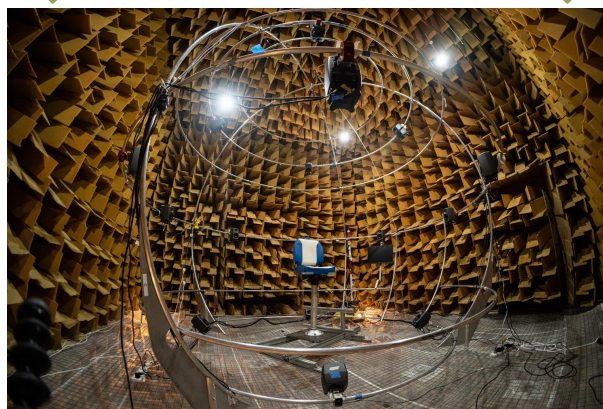


Microphone array used for
acoustics measurement

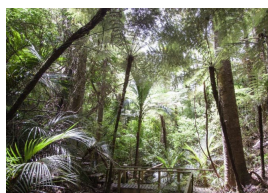


<https://en.wikipedia.org/wiki/Orchestra>

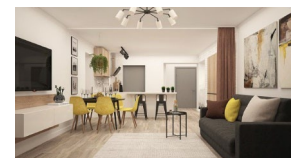
Concert Hall



Loudspeaker array in
anechoic chamber



Outdoor



Domestic
environment

▶ 50

MECHENG726 2026 Chap5 Microphones
& Loudspeakers